webSPDZ · ATLAS demo — Docker quick start

Run the 4-party Surplus Matching demo with one command. No Emscripten, no Python/Node setup, no manual certificate wrangling. Works the same on Linux, Intel Mac, and Apple Silicon Mac.

1 What you need installed

- **Docker Desktop** (Mac or Windows) or **Docker Engine + Compose** (Linux) — that's it. No Python, Node, or Emscripten required on your machine.
- Google Chrome (recommended — this demo has been tested there).

2 Get the project

Unzip or clone the `webSPDZ` folder you were sent, then open a terminal in it:

```
cd webSPDZ
```

3 Start the demo

```
docker compose up --build
```

First run takes under a minute (it's just packaging two small servers — the WebAssembly program is already built into the image). Leave this terminal open; it's running both servers:

- `web` — serves the page over HTTPS on port `8000`
- `signaling` — WebRTC signaling server (WebSocket) on port `8080`

Stop everything later with `Ctrl+C`, or `docker compose down` from another terminal.

4 Trust the certificate (first time only)

This demo uses a self-signed HTTPS certificate for local testing. Chrome will warn you — that's expected, not a sign anything's broken.

Visit each of these once and click through the warning (*Advanced → Proceed to localhost*):

```
https://localhost:8000
https://localhost:8080
```

Both need to be trusted — the signaling server also speaks `wss://`, which Chrome blocks if its certificate isn't accepted.

5 Open the 4 players

Open these 4 URLs in 4 separate Chrome tabs, one per participant — all on this same computer:

P0 `https://localhost:8000/atlas-party.html?arguments=-N,4,-T,1,-w,0,0,surplus_matching_private,-OF,.`

P1 `https://localhost:8000/atlas-party.html?arguments=-N,4,-T,1,-w,0,1,surplus_matching_private,-OF,.`

P2 `https://localhost:8000/atlas-party.html?arguments=-N,4,-T,1,-w,0,2,surplus_matching_private,-OF,.`

P3 `https://localhost:8000/atlas-party.html?arguments=-N,4,-T,1,-w,0,3,surplus_matching_private,-OF,.`

Copy these carefully. Browsers sometimes crop the last characters when you copy/paste a long URL — every link above must end exactly in `-OF,.` (comma, then a literal period). If a tab shows a wall of help text and `ERROR: incorrect number of arguments`, that's this — check nothing got cut off the end.

Don't close any tab until all 4 finish. Each one is waiting for the other 3 to connect before the computation starts.

6 Reading the result

Each tab shows its own result privately — **YOUR PARTNER CODE = …** (`-1` means no useful match was found for that participant). The run should end with **Finished**. Status lines in red (`Using statistical security parameter 40`, `All clients connected!`, etc.) are normal — Emscripten prints its status messages to `console.error`, Chrome just colors that red.

7 Troubleshooting

Symptom	What it means
`port is already allocated` when starting	Something else on your machine is already using 8000 or 8080. Stop it, or edit the `ports:` lines in `docker-compose.yml` to e.g. `18000:8000` (and adjust the URLs you open to match).
Page loads but tabs never connect / stay on "Connect"	The certificate on `:8080` (the signaling server) wasn't accepted yet — revisit `https://localhost:8080` directly and click through the warning.
`Uncaught [object WebAssembly.Exception]` in the console	Almost always a wrong/truncated `arguments=` value — check the program name is exactly `surplus_matching_private` and the URL isn't cut off.
Want different input numbers for the 4 players	Not possible from this Docker image alone — the demo's input values are baked into the WebAssembly build at compile time. That needs a full rebuild with the Emscripten toolchain on a dev machine (see the companion setup guide), then rebuilding this image.

8 What this image does *not* do

It only serves the already-compiled demo exactly as built — it does not install Emscripten or recompile anything. That keeps it small and identical across machines, but it means the 4 players' inputs are fixed to whatever sample values were baked in when the image was built.